

SUSMIJA REDDY JABBIREDDY

✉ jsreddy@umd.edu ☎ (443) 877-2631 🔗 linkedin.com/in/susmija/

EDUCATION

University of Maryland, College Park

Ph.D. Candidate in Computer Science

Advisor: Prof. Amitabh Varshney

May 2024

College Park, MD

GPA: 4.0/4.0

Indian Institute of Technology, Kharagpur

Master of Technology in Computer Science & Engineering

Bachelor of Technology (Honors) in Computer Science & Engineering

Advisor: Prof. Sourangshu Bhattacharya

May 2017

West Bengal, India

GPA: 9.3/10

SKILL SET

Research

Computer Vision, Computer Graphics, Machine Learning, Deep Learning, AR/VR

Languages and Frameworks

Python, C, C++, Java, MATLAB, Pytorch, Tensorflow, OpenGL

Coursework

Deep Learning, Advanced Numerical Optimization, Computational Linguistics, Image Processing,

Advanced Techniques in Visual Learning and Recognition, Foundations of Deep Learning,

Advanced Computer Graphics, Machine Learning, Computational Imaging

PUBLICATIONS AND PATENTS

Towards Sparse Invariant Layout of Optical Antennas for Dynamic Holographic Displays

Susmija Jabbireddy, Yang Zhang, Mario Dagenias, Christopher Metzler, Martin Peckerar, Amitabh Varshney

Under submission

Accelerated Volume Rendering with Volume Guided Neural Denoising

Susmija Jabbireddy, Shuo Li, Xiaoxu Meng, Judith Terrill, and Amitabh Varshney

EuroGraphics Conference on Visualization 2023

VIINTER: View Interpolation With Implicit Neural Representations of Images

Brandon Y. Feng, *Susmija Jabbireddy*, Amitabh Varshney

SIGGRAPH Asia 2022

Sparse Nanophotonic Phased Arrays for Energy-Efficient Holographic Displays

Susmija Jabbireddy, Yang Zhang, Martin Peckerar, Mario Dagenias, Amitabh Varshney

IEEE Conference on Virtual Reality and 3D User Interfaces (VR) 2022

Rectangular Mapping-based Foveated Rendering

Jiannan Ye, Anqi Xie, *Susmija Jabbireddy*, Yunchuan Li, Xubo Yang, Xiaoxu Meng

IEEE Conference on Virtual Reality and 3D User Interfaces (VR) 2022

Foveated Rendering: Motivation, Taxonomy, and Research Directions

Susmija Jabbireddy, Xuetong Sun, Xiaoxu Meng, Amitabh Varshney

arXiv Preprint 2022

Improved Modeling of 3D Shapes with Multi-view Depth Maps

Kamal Gupta*, *Susmija Jabbireddy**, Ketul Shah*, Abhinav Shrivastava, Matthias Zwicker

Oral International Conference on 3D Vision 2020

Task-Specific Representation Learning for Web-Scale Entity Disambiguation

Rijula Kar, *Susmija Reddy*, Sourangshu Bhattacharya, Anirban Dasgupta, Soumen Chakrabarti

Thirty-Second AAAI Conference on Artificial Intelligence, 2018

Extraction of medically interpretable features for classification of malignancy in breast thermography

Himanshu Madhu, Siva Teja Kakileti, Krithika Venkataramani, *Susmija Jabbireddy*

Thirty-Eighth Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2016

Thermography-based breast cancer screening using a measure of symmetry

Krithika Venkataramani, *Susmija Jabbireddy*, Himanshu J. Madhu, Siva Teja Kakileti, Hadonahalli V. Ramprakash

U.S. Patent 10,307,141

Contour-based determination of malignant tissue in a thermal image

Krithika Venkataramani, *Susmija Jabbireddy*, Himanshu J. Madhu, Siva Teja Kakileti

U.S. Patent 9,865,052

PROFESSIONAL EXPERIENCE

Google, Mountain View, California

Software Engineer Intern, Geo 3D

June 2023 – Aug 2023

Digital Terrain Map Estimation

Host: Brett Allen, Software Engineer, Google

- Developed an end-to-end model to estimate terrain from aerial multi-view stereo images.

Google, Mountain View, California
Software Engineer Intern, Geo Pose

June 2022 – Aug 2022

Aerial Pose Evaluation

Host: Daniya Zamalieva, Software Engineer, Google

- Developed production level code for aerial pose evaluation system

Mitsubishi Electric Research Laboratories (MERL), Cambridge, Massachusetts
Research Intern

May 2021 – May 2022

Exploring Implicit Neural Representations

Host: Michael J. Jones, Senior Principal Research Scientist, MERL

- Explored implicit neural representations for downstream computer vision tasks

Sprinklr, Gurugram, India
Product Engineer, Machine Learning

July 2017 – June 2018

Scene Text Detection and Recognition

- Experimented with a two step network - CTPN followed by CRNN to detect text from natural images
- Integrated a single network for detecting and recognizing text into Sprinklr framework with high throughput and efficiency

Brand Logo Detection using Deep Learning

- Integrated FasterRCNN into Sprinklr Framework for recognizing and localizing logos in natural images
- Implemented several data augmentation techniques and filtered noisy images for training

Logo Detection and Recognition

- Worked on logo recognition techniques using visual features from a small number of reference images.
- Implemented an algorithm to precisely identify the location of the logo within the image

Xerox Research Center India, Bangalore, India
Research Intern, Health Care Analytics

May 2016 – July 2016

May 2015 – July 2015

Classifying Thermographic features for Breast Cancer Screening

Host: Krithika Venkatarmani, Visiting Scientist, Xerox Research Center India

- Designed an algorithm to automatically classify thermographic images for the presence of breast cancer
- Automated detection of suspected malignant locations in different views of the breast image
- Additionally attempted to distinguish between various breast conditions

ACADEMIC EXPERIENCE

Research Assistant - University of Maryland, College Park
Towards Leveraging Sparsity for Immersive and Interactive 3D Displays

Jan 2019 – Present

Advisor: Prof. Amitabh Varshney

- Working towards developing fast and efficient algorithms for Computer Generated Holograms (CGH) for VR/AR displays
- Investigating computational approaches to enable the development of efficient large-scale holographic displays

M.Tech Thesis - Indian Institute of Technology, Kharagpur
Representation Learning and Random Projections for Sparse Data

July 2016 – April 2017

Advisor: Prof. Sourangshu Bhattacharya

- Experimented with various hashing techniques to achieve dimensionality reduction
- Investigated multi-task representation learning (MTRL) for Named Entity Disambiguation (NED)

B.Tech Project - Indian Institute of Technology, Kharagpur
Distributed Matrix Operations using Spark

July 2015 – April 2016

Advisor: Prof. Sourangshu Bhattacharya

- Designed an algorithm to compute the inverse of a high dimension symmetric positive definite matrix
- Implemented the algorithm in Apache Spark in a distributed environment using HDFS for data storage

Detecting Reflective Surfaces from a Single Image

Mentor: Prof. Abhinav Shrivatsava

Aug 2019 – Dec 2019
University of Maryland

- Worked on detecting objects with reflective surfaces using depth discontinuity information

SCHOLASTIC ACHIEVEMENTS

- Finalist for **UMD Innovation of the Year Award**, 2023 for work on Sparse Nanophotonic Arrays for Holographic Displays
- Received the IEEE VR Inclusion, Diversity, and Accessibility Scholarship 2022
- Selected for attending CRA Grad Cohort for Women 2019, 2020
- Recipient of the prestigious University of Maryland, College Park **Dean's Fellowship** for the academic years 2018–2020
- Achieved **Best Master's Thesis Project Award** in IIT Kharagpur for the academic year 2016–2017